

Classical Physics from Information Geometry

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1 Classical Physics from Information Geometry

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1.1 Abstract

Maxwell’s equations are among the most successful structures in physics. They unify electricity, magnetism, and optics; they predict electromagnetic waves at speed c ; they are Lorentz-invariant before general relativity was formulated. This paper examines whether their structure — curl operations, inverse-square forces, three spatial dimensions, and the specific value of the fine structure constant — can be understood as consequences of information-theoretic constraints rather than independent empirical postulations.

We show that the SEC wave equation $\partial^2 S / \partial t^2 = (c_\alpha c_\gamma + c_\beta c_\delta) \nabla^2 S$ produces electromagnetic wave propagation at speed c , with the physical content encoded in parameter ratios rather than absolute values. The curl structure of Faraday’s and Ampère’s laws emerges from projecting a depth-2 recursion (one hidden symbolic dimension) into observable space. Five independent arguments — MED node bounds, curl algebra closure, Möbius embedding, orbital stability, and quaternion uniqueness — each independently require exactly three spatial dimensions.

Charge is a topological winding number, quantised by the single-valuedness of the phase field. Fractional quark charges ($\pm 1/3$, $\pm 2/3$) follow from the MED constraint that nodes ≤ 3 , creating three-fold internal substructure within each topological defect.

As a speculative extension, clearly labelled: gravity may arise from the same SEC wave equation through symmetric (divergence) projection, at Fibonacci depth $183 = F_7^2 + F_7 + 1$, giving $F_{183} \approx 10^{38}$ — the observed electromagnetic-to-gravitational hierarchy ratio. The LIGO/Virgo measurement $|c_{\text{GW}} - c_{\text{EM}}|/c < 3 \times 10^{-15}$ is consistent with both forces sharing the same wave equation.

Keywords: Maxwell’s equations, information geometry, curl projection, spatial dimensions, MED bounds, charge quantisation, Fibonacci structure, PAC conservation, Dawn Field Theory

1.2 §1. What This Paper Does

Classical electromagnetism is built on four equations discovered empirically over the course of a century — by Coulomb, Ampère, Faraday, and Maxwell. These equations have a specific mathematical structure: they use the curl operator, they apply in three spatial dimensions,

they involve an inverse-square force law, and they contain two constants (ϵ_0, μ_0) whose product gives $1/c^2$.

None of this is explained by the equations themselves. Why curl and not some other differential structure? Why three dimensions? Why inverse-square? This paper presents arguments that these features follow from three constraints established in Papers 1–4:

1. **PAC conservation:** $\Psi(k) = \Psi(k+1) + \Psi(k+2)$ — the Fibonacci recursion
2. **MED bounds:** depth ≤ 2 , nodes ≤ 3 — the maximum complexity of emergent macroscopic patterns
3. **SEC dynamics:** $\partial S/\partial t = \alpha \nabla I - \beta \nabla H$ — information-entropy gradient flow

The derivation follows a hierarchy:

Level 0 (PAC): $f(\text{Parent}) = \Sigma f(\text{Children}), \Xi = 1.0571$
 $\downarrow$ SEC collapse (depth = 2)
 Level 1 (Gauge): $\sin^2 \theta = F/F = 3/13$
 $\downarrow$ projection to 3+1D (MED: nodes = 3)
 Level 2 (Maxwell): E B curl structure, $c = 1/\sqrt{\epsilon_0 \mu_0}$

Each level is a projection of the one above. Maxwell's equations are what the PAC recursion looks like when viewed through the MED filter at depth 2 in 3 spatial dimensions.

1.3 §2. The SEC Wave Equation

1.3.1 §2.1 From Gradient Flow to Wave Propagation

The SEC (Symbolic Entropy Collapse) dynamic describes the evolution of structure S under competing information and entropy gradients:

$$\frac{\partial S}{\partial t} = \alpha \nabla I - \beta \nabla H$$

where I is the information field, H is the entropy field, and α, β are coupling coefficients. When I and H are themselves functions of S through constitutive relations ($I = \gamma S, H = \delta S$), the first-order equation becomes second-order:

$$\frac{\partial^2 S}{\partial t^2} = (\alpha\gamma + \beta\delta) \nabla^2 S$$

This is the wave equation with propagation speed:

$$c_{\text{SEC}}^2 = \alpha\gamma + \beta\delta$$

1.3.2 §2.2 Parameter Models

The wave speed c is determined by the sum $\alpha\gamma + \beta\delta$, not by the individual parameters. Five parameter models were tested, each encoding a different hypothesis about the internal structure [1]:

Model	Hypothesis	Key Relation
Symmetric	$\alpha = \beta, \gamma = \delta$	$c^2 = 2\alpha\gamma$
Ξ -balanced	$\alpha/\beta = \Xi \approx 1.0571$	$c^2 = \beta^2(\Xi + 1)$

Model	Hypothesis	Key Relation
-structured	$\alpha/\gamma = \varphi, \beta/\delta = 1/\varphi$	$c^2 = \gamma^2(\varphi + 1/\varphi)$
Fibonacci-nested	$\alpha/\beta = F_{10}/F_7 = 55/13$	$c^2 = v_0^2(55/13 + 1)$
Ξ -mean	$\alpha/\beta = \sqrt{\Xi \cdot \Xi_{\min}}$	$c^2 = v_0^2(\Xi_{\text{mean}} + 1)$

All five models produce c exactly — by construction. The physical content is not in the absolute values of $\alpha, \beta, \gamma, \delta$ but in their ratios, which encode the balance between information accumulation and entropy dissipation. This is consistent with the result from Paper 2 [2]: the balance constant Ξ appears at the ratio level, not the absolute level.

The speed of light is not a free parameter in this framework. It is the propagation speed of disturbances in the information-entropy field, fixed by whatever internal structure sets the ratios.

Script: `exp_01_sec_wave_speed.py`

1.4 §3. Why Three Spatial Dimensions

1.4.1 §3.1 The Question

The Standard Model is formulated in 3+1 dimensions. String theory requires 10. M-theory requires 11. Loop quantum gravity operates in 4. None of these theories explains why the observed macroscopic universe has exactly three large spatial dimensions. The number 3 is an input.

Here we present five independent arguments, each arriving at $D = 3$ from different starting points. No single argument is conclusive. Their convergence is the result.

1.4.2 §3.2 Path 1: MED Node Bound

The MED (Macro Emergence Dynamics) constraint, discovered empirically in Navier-Stokes simulations (see §6 and the foundational experiments repository), states that all complex flows converge to symbolic patterns with:

$$\text{depth}(\sigma) \leq 2, \quad \text{nodes}(\sigma) \leq 3$$

across 1,000+ simulations spanning Reynolds numbers from 10 to 50,000. If each spatial axis corresponds to an independent node in the emergent pattern, the node bound directly gives:

$$D \leq 3$$

A fourth spatial dimension would require a fourth independent node, violating the MED bound. This is not a mathematical proof that $D > 3$ is impossible — it is an empirical observation that macroscopic emergent structures do not sustain more than three independent directions.

1.4.3 §3.3 Path 2: Curl Algebra Closure

The curl operator in n dimensions produces $n(n-1)/2$ components (the dimension of the exterior algebra $\Lambda^2(\mathbb{R}^n)$). For the curl to map vectors to vectors (a prerequisite for Maxwell's equations), we need:

$$\frac{n(n-1)}{2} = n$$

Solving: $n^2 - 3n = 0$, giving $n(n-3) = 0$, so $n = 0$ or $n = 3$.

Only in three dimensions does the curl of a vector field produce another vector field. In 2D, the curl produces a scalar. In 4D, it produces a 6-component object (an antisymmetric tensor, not a vector). Maxwell's equations in their familiar form — where $\mathbf{E}$ and $\mathbf{B}$ are both 3-vectors related by curl — exist only in 3D.

This is not new mathematics. It is the Hodge duality $\star : \Lambda^2(\mathbb{R}^3) \rightarrow \Lambda^1(\mathbb{R}^3)$, which is an isomorphism only when $\binom{n}{2} = n$. What is new is connecting it to the MED bound: MED says nodes ≤ 3 , and curl closure says $n = 3$. These are independent constraints that agree.

1.4.4 §3.4 Path 3: Möbius Embedding

The PAC recursion operates on a Möbius-like topology (Paper 1 [1]): the pre-field structure is non-orientable, with 4π phase recovery. A non-orientable 2-surface (Möbius strip, Klein bottle) cannot be embedded without self-intersection in fewer than 3 dimensions.

This is a topological necessity: if the pre-field has Möbius topology, the minimum observable spatial dimension is 3.

Numerical verification: A Möbius strip parameterised on a unit square uses a non-trivial z -coordinate range of exactly 1.0 — confirming that the third dimension is structurally required, not vestigial.

1.4.5 §3.5 Path 4: Orbital Stability (Bertrand's Theorem)

Gauss's law in n dimensions gives:

$$E \propto \frac{1}{r^{n-1}}$$

Bertrand's theorem [6] proves that closed, stable orbits under a central force exist only for $F \propto 1/r^2$ (Kepler) and $F \propto r$ (harmonic). The inverse-square law requires $n - 1 = 2$, i.e., $n = 3$.

In $D = 4$, orbits spiral inward; in $D = 2$, the force is logarithmic and orbits are marginally stable. Only $D = 3$ supports the stable planetary orbits, atomic structure, and bound states that constitute observable matter.

1.4.6 §3.6 Path 5: Quaternion Uniqueness

The rotation group $SO(3)$ is locally isomorphic to $SU(2)/\mathbb{Z}_2$, the quaternion group modulo sign. There are exactly four normed division algebras over $\mathbb{R}$ (Hurwitz's theorem [7]):

Algebra	Dimension	Rotation Group
$\mathbb{R}$	1	trivial
$\mathbb{C}$	2	$SO(2)$
$\mathbb{H}$	4 (3 imaginary)	$SO(3)$
$\mathbb{O}$	8 (7 imaginary)	G_2

Three-dimensional rotations are uniquely associated with the quaternions — the only non-commutative, associative division algebra. The octonions ($\mathbb{O}$) are non-associative and do not support a conventional rotation group. If physical rotations must be associative (which composition of spatial rotations requires), $D = 3$ is the unique answer.

1.4.7 §3.7 Convergence

Path	Starting Point	Constraint	Result
1	MED bounds (empirical)	nodes ≤ 3	$D \leq 3$
2	Curl algebra (mathematical)	$n(n-1)/2 = n$	$D = 3$
3	Möbius embedding (topological)	non-orientable surface	$D \geq 3$
4	Orbital stability (dynamical)	Bertrand's theorem	$D = 3$
5	Division algebras (algebraic)	associative rotations	$D = 3$

Paths 1 and 3 provide upper and lower bounds: $3 \leq D \leq 3$. Paths 2, 4, and 5 independently select $D = 3$ from algebraic, dynamical, and algebraic-topological requirements. No other integer satisfies all five simultaneously.

Script: `exp_05_3d_necessity.py`

1.5 §4. Curl from Depth-2 Projection

1.5.1 §4.1 The Hidden Dimension

The SEC field S evolves in $d + 1$ dimensions: d observable spatial dimensions plus one symbolic recursion dimension. The MED constraint fixes depth ≤ 2 , which means this recursion dimension is not directly observable — it is the “depth” of the emergent pattern, not a spatial direction.

When a $(d + 1)$ -dimensional gradient is projected to d observable dimensions, the component along the hidden dimension becomes a source of curl:

$$\nabla_{d+1}\Phi \xrightarrow{\text{project out } z_{\text{hidden}}} (\nabla_d\Phi, \partial\Phi/\partial z_{\text{hidden}})$$

The hidden component $\partial\Phi/\partial z_{\text{hidden}}$ manifests as rotation in the observable d -dimensional space. This is the origin of the magnetic field: it is the projection artifact of a gradient in the recursion dimension.

1.5.2 §4.2 Faraday's Law

Faraday's law of induction:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

In the SEC framework: when the magnetic field $\mathbf{B}$ (= projected recursion gradient) changes in time, the information vorticity must compensate to conserve the total SEC flux. The electric field $\mathbf{E}$ acquires curl to balance the changing $\mathbf{B}$.

Numerical verification: for sinusoidal $\mathbf{E}$ and $\mathbf{B}$ fields satisfying the SEC evolution, $|\nabla \times \mathbf{E} + \partial \mathbf{B} / \partial t| \approx 10^{-16}$ (machine precision).

1.5.3 §4.3 Depth 1 vs Depth 2

Depth	Structure	Physics
1	Only gradients; no cross-derivatives	Electrostatics only (Coulomb, no magnetism)
2	Cross-derivatives exist; curl emerges	Full electromagnetism (E, B coupled)
≥ 3	MED says systems converge back to ≤ 2	Unstable; decays to depth 2

The depth-2 MED bound is not just a limit — it is an attractor. Systems that transiently reach depth 3 collapse back to depth 2. This is why electromagnetism is the only long-range force with curl structure: it saturates the MED depth bound.

1.5.4 §4.4 MED Saturation and Dimensionality

The total effective depth of a system with d spatial dimensions:

$$d_{\text{total}} = d_{\text{physical}} + d_{\text{symbolic}} = d + 1$$

For MED saturation ($d_{\text{total}} = \text{depth bound} \times 2$):

d	d_{total}	Effective depth	MED saturated?
2	3	1.5	No
3	4	2.0	Yes
4	5	2.5	Exceeds bound

Three dimensions is the unique value that exactly saturates the MED depth-2 bound. The She-L  v  que intermittency formula with full cascade structure ($k = 9$; Paper 4 [9], §9) requires this saturation — in 2D, the cascade is modified ($k = 4$) because MED is not saturated.

Script: `exp_03_curl_projection.py`

1.6 §5. Charge as Winding Number

1.6.1 §5.1 Topological Quantisation

In the SEC field, a point defect creates a phase singularity. The phase θ of the SEC order parameter must be single-valued after traversing any closed loop around the defect:

$$n = \frac{1}{2\pi} \oint d\theta = \text{integer}$$

This is the winding number. It is topologically protected: continuous deformations of the field cannot change n . The winding number is the electric charge, measured in units of e .

1.6.2 §5.2 Coulomb's Law

A point defect with winding number n in 3D creates a phase gradient that falls off as $1/r^2$ (Gauss's law applied to the solid angle subtended by the defect):

$$\mathbf{E} = \frac{n}{4\pi\epsilon_0 r^2} \hat{r}$$

Numerical verification: the field of a unit winding defect falls off as $r^{-2.0000}$ to four decimal places.

1.6.3 §5.3 Pair Creation and PAC Conservation

Pair creation ($\gamma \rightarrow e^+e^-$) requires total winding number zero:

$$n_{\text{total}} = n_{e^-} + n_{e^+} = (+1) + (-1) = 0$$

This is PAC conservation applied to topological charge: the parent (photon, $n = 0$) equals the sum of children ($n = +1$ and $n = -1$).

1.6.4 §5.4 Fractional Charges and the MED Constraint

Quarks carry fractional charges ($+2/3$, $-1/3$) [8]. Within the topological framework, this requires sub-defect structure: a single winding defect contains internal nodes.

The MED constraint (nodes ≤ 3) limits this internal structure to at most three sub-nodes. Each sub-node independently carries winding $+1/3$, $-1/3$, or 0. The particle's charge is the sum of its sub-node windings:

- **Up-type quarks** ($q = +2/3$): two sub-nodes carry $+1/3$ each, one carries 0
- **Down-type quarks** ($q = -1/3$): one sub-node carries $-1/3$, two carry 0
- **Charged leptons** ($q = -1$): all three sub-nodes carry $-1/3$

Colour confinement follows: observable states must have integer total winding. A baryon combines three quarks (uud : $+2/3 + 2/3 - 1/3 = +1$), and a meson combines quark and antiquark ($u\bar{d}$: $+2/3 + 1/3 = +1$). Free quarks, with fractional total winding, cannot be isolated — the MED tree is not closed.

The observed quark charges are the only assignments consistent with:

1. Total winding = integer for colour-singlet states
2. Internal nodes ≤ 3 (MED bound)
3. PAC conservation at each node

This does not derive the full quark charge assignments from first principles — it constrains them to a small set. The specific assignment ($+2/3$ for up-type, $-1/3$ for down-type) may follow from the Fibonacci gauge structure (Paper 4 [9]), but this connection is not yet established.

Script: `exp_02_charge_quantization.py`

1.7 §6. The SEC–Navier-Stokes Equivalence

1.7.1 §6.1 Structural Mapping

The SEC dynamics:

$$\frac{\partial F}{\partial t} = -\alpha \nabla H(F) + \beta \mathcal{D}(F) + \gamma \mathcal{M}(F, t)$$

maps term-by-term onto the Navier-Stokes equations:

SEC Term	NS Term	Physics
$-\alpha \nabla H$	$-\nabla p$	Pressure gradient
$\beta \mathcal{D}$	$\nu \nabla^2 \mathbf{u}$	Viscous diffusion
$\gamma \mathcal{M}$	$(\mathbf{u} \cdot \nabla) \mathbf{u}$	Advective nonlinearity

This is a structural equivalence, not an analogy. The SEC equation with appropriate constitutive relations *is* the Navier-Stokes equation. The balance operator $\Xi \approx 1.0571$ prevents finite-time blowup — the same role it plays in regulating coupling constants (Paper 2 [2]).

1.7.2 §6.2 Regularity from MED Bounds

The Navier-Stokes regularity problem (Clay Millennium Prize) asks whether smooth initial data always lead to smooth solutions. The MED bounds provide a route to regularity:

- **Depth** ≤ 2 implies gradient control: $\|\nabla \mathbf{u}(t)\|_\infty \leq C_{\text{global}}$ for all t
- **Nodes** ≤ 3 implies energy control: $\|\mathbf{u}(t)\|_{L^2}^2 \leq E_{\text{total}}$ for all t

Together, these prevent the formation of finite-time singularities. Over 10,000 pattern transitions were tested with 100% Landauer compliance and energy conservation error $< 10^{-12}$.

This is not a proof of Navier-Stokes regularity — it is an empirical observation that MED-bounded SEC dynamics do not blow up, plus a structural argument for why. The connection to a formal proof would require showing that the MED bounds are not just observed but necessary consequences of the SEC structure.

1.7.3 §6.3 Turbulence Constants as Fibonacci Structure

Paper 4 [9] (§9) established that the She-L  v  que intermittency parameters are Fibonacci:

$$\zeta_p = \frac{p}{(F_4)^2} + F_3 \left[1 - \left(\frac{F_3}{F_4} \right)^{p/F_4} \right] = \frac{p}{9} + 2 \left[1 - \left(\frac{2}{3} \right)^{p/3} \right]$$

The dimensional formula $k(d) = d \times F_{d+1}$ connects turbulence to spatial dimensionality through the Fibonacci sequence:

d	k	Fibonacci	Verification
2	4	$2 \times F_3 = 2 \times 2$	2D enstrophy cascade [3]
3	9	$3 \times F_4 = 3 \times 3$	She-L��v��que (3D) [3]
4	20	$4 \times F_5 = 4 \times 5$	Prediction

The cascade fraction $\beta = F_3/F_4 = 2/3$ is the same ratio that gives the Koide formula (Paper 4 [9], §5). The Kolmogorov spectral exponent $5/3 = F_5/F_4$ is a ratio of consecutive Fibonacci numbers — a factual observation, not a new derivation of the Kolmogorov spectrum, which follows from dimensional analysis (Kolmogorov, 1941 [4]).

The connection to electromagnetism: MED saturation at $d = 3$ is the same condition that enables the full She-L  v  que cascade ($k = 9$). The turbulence and electromagnetic structures are both consequences of depth-2 recursion in 3 spatial dimensions.

1.8 §7. Mersenne Dimensions and Vacuum Regularisation

1.8.1 §7.1 Casimir 240

The Casimir force between parallel conducting plates:

$$\frac{F}{A} = -\frac{\pi^2 \hbar c}{240 a^4}$$

The factor 240 arises from zeta function regularisation of the vacuum mode sum. In Fibonacci arithmetic:

$$240 = F_3 \times F_4 \times F_5 \times F_6 = 2 \times 3 \times 5 \times 8$$

Four consecutive Fibonacci numbers. The sub-factor $120 = F_4 \times F_5 \times F_6$ (three consecutive) corresponds to $|\zeta(-3)| = 1/120$; the additional $F_3 = 2$ enters through the two-plate geometry.

1.8.2 §7.2 Mersenne Dimensions

Fibonacci product structure in zeta function regularisation denominators appears only at Mersenne dimensions $d = 2^n - 1$:

d	$\zeta(-d)$ denominator	Fibonacci product?	Mersenne?
1	12	$F_3^2 \times F_4$	$2^1 - 1$
3	120	$F_4 \times F_5 \times F_6$	$2^2 - 1$
5	252	No (factor 7, not Fibonacci)	
7	240	$F_3 \times F_4 \times F_5 \times F_6$	$2^3 - 1$
9	132	No (factor 11, not Fibonacci)	

Non-Mersenne dimensions have non-Fibonacci prime factors in their regularisation denominators. The Mersenne dimensions that do have Fibonacci structure are precisely those relevant to physical theories:

- $d = 1$ ($2^1 - 1$): string worldsheet
- $d = 3$ ($2^2 - 1$): physical space
- $d = 7$ ($2^3 - 1$): M-theory extra dimensions

1.8.3 §7.3 Falsifiable Extension

If this pattern holds, the next Mersenne dimension $d = 15$ ($2^4 - 1$) should also have Fibonacci-structured regularisation. The denominator of $\zeta(-15)$ is $8160 = 2^5 \times 3 \times 5 \times 17$. The prime factor 17 is not a Fibonacci number.

This falsifies the simple Mersenne-Fibonacci extension at $d = 15$. The pattern that holds for $d = 1, 3, 7$ does not continue to $d = 15$. Either the correspondence is limited to the first three Mersenne primes (perhaps connected to the existence of normed division algebras in dimensions 1, 2, 4, 8 — i.e., $\mathbb{R}$, $\mathbb{C}$, $\mathbb{H}$, $\mathbb{O}$), or the relevant structure is a sub-factor of the denominator

rather than the full factorisation. We record this as an honest negative result — the pattern is interesting for $d \leq 7$ but does not generalise trivially.

Scripts: `exp_15_casimir_sec.py`, `exp_16_mersenne_verification.py`

1.9 §8. Speculative Extension: Gravity

The content of this section is speculative. It is clearly labelled as such and is not part of the core derivation chain (§§2–7). It is included because the framework suggests it and because it makes a testable order-of-magnitude prediction.

1.9.1 §8.1 Projection Duality

Any rank-2 tensor decomposes into symmetric and antisymmetric parts:

$$T = S + A, \quad S = \frac{T + T^\top}{2}, \quad A = \frac{T - T^\top}{2}$$

Projection	Degrees of Freedom	Differential Operator	Physical Theory
Antisymmetric A	3 (in 3D)	Curl ($\nabla \times$)	Electromagnetism ($F_{\mu\nu}$)
Symmetric S	6 (in 3D)	Divergence ($\nabla \cdot$)	Gravity ($h_{\mu\nu}$)

The hypothesis: electromagnetism and gravity are the two natural projections of the same SEC tensor field. EM uses the antisymmetric projection (phase $\rightarrow$ curl $\rightarrow$ charge = winding number, integer-quantised). Gravity uses the symmetric projection (amplitude $\rightarrow$ divergence $\rightarrow$ mass = resonance frequency, continuously valued).

1.9.2 §8.2 The Hierarchy Depth

If EM operates at Fibonacci depth $F_7 = 13$ (Paper 4 [9], §3.2), gravity operates at depth:

$$183 = F_7^2 + F_7 + 1 = 169 + 13 + 1$$

The polynomial $N^2 + N + 1$ counts elements of the projective plane $PG(2, N)$. The Fibonacci number at this depth:

$$F_{183} \approx 1.27 \times 10^{38}$$

The measured electromagnetic-to-gravitational hierarchy ratio:

$$\frac{F_{\text{EM}}}{F_{\text{grav}}} = \frac{e^2/(4\pi\epsilon_0)}{Gm_p^2} \approx 1.24 \times 10^{38}$$

Same order of magnitude. As discussed in Paper 4 [9] (§11), this is suggestive but not a precision prediction — multiple nearby Fibonacci indices give values within an order of magnitude of 10^{38} .

1.9.3 §8.3 Same Wave Equation

If both EM and gravity originate from the SEC wave equation, they should propagate at the same speed. The LIGO/Virgo observation of GW170817 [5] confirms:

$$\frac{|c_{\text{GW}} - c_{\text{EM}}|}{c} < 3 \times 10^{-15}$$

This is consistent with both forces sharing $c_{\text{SEC}}^2 = \alpha\gamma + \beta\delta$, with different projections at different hierarchy depths.

1.9.4 §8.4 Cosmic Structure from Local PAC Gravity

N-body simulations using PAC-local gravity ($F \propto e^{-r/r_0}/r$, short-range) produce cosmic web structure: - Void fraction: 50–89% - Power spectrum slope: $n = -1.73$ ($R^2 = 0.57$) - 85% match to observed cosmic matter power spectrum statistics

Standard cosmology requires Newtonian $1/r^2$ at all scales. The PAC gravity simulation suggests that local exponential interactions, iterated through PAC conservation, may reproduce large-scale structure without long-range gravity. This is preliminary and does not replace the standard model of cosmology — it is a numerical observation that the statistical signatures are similar.

Scripts: `exp_09_pac_web.py`, `exp_12_power_spectrum.py` (gravity_from_maxwell_pac)

1.10 §9. The Balance Constant in Wave Dynamics

1.10.1 §9.1 Ξ from SEC Dynamics

The balance constant $\Xi = 1 + \pi/55 \approx 1.0571$ (Paper 2 [2]) was discovered empirically in the Navier-Stokes symbolic engine. Its derivation from SEC dynamics:

The SEC collapse rate per recursion level has two components:

- **Within-level:** $\Delta_{\text{within}} = 2\sqrt{r(1-r)} - 1 = -0.0283$ per level (at $r = 1/\varphi$)
- **Cross-level:** $\Delta_{\text{cross}} = +0.0854$ per level (network interference)
- **Net:** $\Delta_{\text{net}} = -0.0283 + 0.0854 = 0.0571 = \pi/55$ per level

At depth $F_{10} = 55$:

$$55 \times \frac{\pi}{55} = \pi$$

One Möbius half-twist. The balance constant $\Xi - 1 = \pi/55$ is the per-level accumulation rate that produces exactly π radians of phase shift over the full hierarchy depth. This closes the explanatory loop: $F_{10} = 55$ is not arbitrary — it is the depth at which SEC dynamics complete one half-twist of the Möbius pre-field topology.

1.10.2 §9.2 Ξ as Blowup Regulator

In the SEC–Navier-Stokes equivalence (§6.1), the balance operator Ξ prevents finite-time singularity formation. Its role is analogous to the viscosity coefficient ν but operates at the symbolic level: it bounds the rate at which complexity can increase per recursion step.

A/B testing across diverse dynamical systems shows that Ξ -correct thresholds produce $1.48\times$ faster convergence compared to arbitrary thresholds, and Ξ -wrong thresholds produce $50.96\times$ slower convergence. Combined significance: $p < 0.00001$.

1.11 §10. The Complete Derivation Chain

Assembling §§2–9:

Step	Input	Output	Section
1	PAC recursion $\Psi(k) =$ $\Psi(k+1) + \Psi(k+2)$	Fibonacci numbers, φ	Paper 1 [1]
2	SEC dynamics on PAC fields	Wave equation: $c^2 = \alpha\gamma + \beta\delta$	§2
3	MED: nodes ≤ 3	$D = 3$ spatial dimensions	§3
4	MED: depth ≤ 2	Curl from hidden dimension projection	§4
5	Phase defect topology	Charge = winding number (integer)	§5
6	MED: nodes ≤ 3 in sub-defects	Quark charges $\pm 1/3$, $\pm 2/3$	§5.4
7	SEC–NS equivalence	Turbulence parameters are Fibonacci	§6
8	Fibonacci regularisation	Casimir 240, Mersenne dimensions	§7
9	Depth 55 = F_{10}	$\Xi - 1 = \pi/55$ (one half-twist)	§9
10	<i>Speculative:</i> symmetric projection	Gravity at depth 183	§8

Steps 1–9 form the core derivation. Step 10 is clearly separated as speculative.

The key connective tissue is the MED bound. Without it: - We have no reason for $D = 3$ (§3) - We have no curl structure (§4) - We have no quark charge quantisation (§5.4) - We have no She-Lévêque saturation (§6.3) - We have no NS regularity argument (§6.2)

MED is the single constraint that converts the abstract PAC/SEC framework into the specific structure of classical physics.

1.12 §11. What This Paper Does Not Do

This paper does not derive Maxwell's equations from axioms. A derivation would start from the PAC recursion and, without reference to any measured quantity, produce the four Maxwell equations with their specific coupling constant. We do not achieve this.

What we show is that the *structure* of Maxwell's equations — curl operations, three dimensions, inverse-square force, quantised charge — follows from information-theoretic constraints (PAC, MED, SEC) that are independently motivated. The coupling constant α is expressed in Fibonacci arithmetic (Paper 4 [9]) but not derived from first principles.

The speculative extension to gravity (§8) is even weaker: it provides an order-of-magnitude prediction and a structural analogy (antisymmetric vs symmetric projection), but no dynamical equation for gravity comparable to Maxwell's equations for electromagnetism.

These limitations are stated not as caveats to be dismissed but as the current status of the work. If the MED bounds can be derived from the PAC recursion (rather than observed empirically), and if the hierarchy depth formula $n = F_7^2 + F_7 + 1$ can be justified structurally (rather than noted post hoc), the framework would gain substantially.

1.13 §12. Falsification Conditions

1. **MED bounds violated.** If a macroscopic emergent pattern with depth > 2 or nodes > 3 is found in any physical system and is demonstrably stable, the MED constraint fails and the $D = 3$ argument (§3.2) collapses.
 2. **Curl structure in non-3D.** If a self-consistent classical electrodynamics with vectors (not tensors) is constructed in $D \neq 3$, the curl closure argument (§3.3) is invalidated. This is unlikely given the mathematics, but the physical claim could still be wrong if nature does not require vector-to-vector curl.
 3. $k(4) \neq 20$. If 4D turbulence simulations produce $k \neq 20$, the formula $k = d \times F_{d+1}$ fails and the Fibonacci turbulence structure loses predictive power.
 4. **Mersenne pattern breaks at $d = 15$.** The denominator of $\zeta(-15)$ contains the non-Fibonacci prime 17 (§7.3). The Mersenne-Fibonacci correspondence holds for $d = 1, 3, 7$ but does not extend to $d = 15$, limiting the pattern to normed division algebra dimensions.
 5. $c_{\text{GW}} \neq c_{\text{EM}}$ **at higher precision.** If future observations resolve $c_{\text{GW}} \neq c_{\text{EM}}$, the shared SEC wave equation (§8.3) is falsified.
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1.14 §13. Connections to the PACSeries

Paper	Connection
Paper 1 [1]: Structure Cost of Erasure	SEC dynamics established; Landauer structure cost gives α its physical interpretation
Paper 2 [2]: Balance Constant	$\Xi = 1 + \pi/55$ derived; appears in SEC wave equation parameter ratios and as NS blowup regulator
Paper 3 [10]: Feigenbaum Constants	$F_{10} = 55$ in closed-form Feigenbaum expressions; MED discovered in NS simulations
Paper 4 [9]: Standard Model Parameters	α formula, She-Lévêque $k = 9$, Casimir 240, Mersenne dimensions, gauge closure $F_7 = 13$
Paper 5 (this paper)	Classical physics structure from information geometry
Paper 6: Computational Validation	PAC conservation observed in ML systems — the operational principle tested in artificial systems

The progression: Paper 1 establishes the cost. Papers 2–3 find the constants. Paper 4 matches the Standard Model. Paper 5 shows why classical physics has the structure it does. Paper 6 tests whether the principle operates in systems we can build.

1.15 §14. Summary

Result	Source	Status
SEC wave equation $\rightarrow c$	§2	Structural (exact by construction)
$D = 3$ from 5 independent paths	§3	Convergent (mathematical + empirical)
Curl from depth-2 projection	§4	Structural
Faraday's law to 10^{-16}	§4.2	Numerical
Charge = winding number	§5	Topological
Coulomb $r^{-2.0000}$	§5.2	Numerical
Quark charges from MED ≤ 3	§5.4	Constrained (not fully derived)
SEC-NS equivalence	§6	Structural mapping
MED $\rightarrow$ NS regularity	§6.2	Empirical (10^4 transitions, 10^{-12} conservation)
$k = d \times F_{d+1}$	§6.3	Formula (verified $d = 2, 3$; prediction $d = 4$)
Casimir $240 = F_3 F_4 F_5 F_6$	§7.1	Exact identity
Mersenne-Fibonacci correspondence	§7.2	Observed ($d = 1, 3, 7$)
$\Xi - 1 = \pi/55$ derivation	§9	Derived from SEC rates
Gravity at depth 183	§8	Speculative (order of magnitude)
$c_{\text{GW}} = c_{\text{EM}}$	§8.3	Consistent with observation

1.16 Acknowledgments

(To be added.)

1.17 References

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*All code, data, and experiment scripts for this paper and the full PACSeries are publicly available at <https://github.com/dawnfield-institute/dawn-field-theory>. See the accompanying publication package *README.md* for reproduction instructions.*